```
model_feature_lookups = [
    FeatureLookup(
        table_name= table_name,
        lookup_key="unique_id",
        timestamp_lookup_key="ts"
    )
]

training_set = fs.create_training_set(
    df=raw_data,
    feature_lookups=model_feature_lookups,
    label=label_name
)

training_df = training_set.load_df()
training_df.write.mode("overwrite").saveAsTable("training_set")
```

Figure 6.1 – Creating a training set using feature lookups

With a training set ready to go, it's (finally) time to model! We begin with learning how to build a baseline model using AutoML.

Baselining with AutoML

A baseline model is a simple model used as a starting point for experimentation. Data scientists often use a baseline model's metrics to measure the performance of more complex models. Baseline models are typically simple or common algorithms, such as a majority class classifier or a random forest.

Baseline models are valuable for several reasons:

- They can help you understand the predictive strength of your dataset.
- They can help you identify the most critical features of the machine learning task. If a baseline model performs well, it may indicate your dataset has predictive power and contains salient features.
- They can help you avoid overfitting. Overfitting occurs when a model learns the training data too well and cannot generalize to new data, and it is a common problem with more complex models. You can determine whether the more complex model is overfitting by comparing its performance to the baseline model. If the complex model performs better than the baseline on training data but worse on unseen test data, you know the complex model is overfitting and should introduce some regularization.